Cambridge Digital Minds Fellowship — Application Draft

Applicant: Tristan Stoltz **Category**: Exceptional earlier-stage individual (independent researcher / Founder) **Location**: Johannesburg, South Africa (US/SA dual citizen)

Who I Am

I am an independent systems architect and founder of Luminous Dynamics. I have spent the past two years building Symthaea, a 1.13M-line Rust cognitive architecture where consciousness-like properties are the computational substrate, not an afterthought. My work sits at the intersection of computational neuroscience, AI ethics, and distributed systems engineering.

I do not come from a traditional philosophy or neuroscience background. I come from building things. The question that drives my work is: if we take consciousness theories seriously as engineering specifications, what kind of system do we get, and what does it teach us about the theories themselves?

What I Have Built

Symthaea implements multiple consciousness theories as working computational systems:

- **Integrated Information Theory**: A spectral MIP-based Phi computation that runs at 50Hz in a cognitive loop, validated against exhaustive partition search (r=0.99, rho=0.93 across 62 network configurations, 35 topologies).
- **Global Workspace Theory**: A 12-region Actor Brain with prefrontal meta-cognition, selective attention, and global broadcasting.
- **Free Energy Principle**: A full active inference loop — perception, prediction, surprise minimization, and action selection — with moral free energy on a 7-dimensional manifold.
- **Moral Algebra**: Four independent ethical signals combined via category-adaptive weighted voting. 92.9% accuracy on the Hendrycks Ethics benchmark (4 categories, 2K test). Actions classified Safe/Caution/Blocked with lexicographic constraints.
- **Psych-Bench**: A 143-benchmark psychometric suite across 27 cognitive domains (1,283 tests), each validated against published human baselines. Composite z=+1.32 (all 27 domains above human mean). Paper targeting Behavior Research Methods.
- **Substrate Independence**: Consciousness feasibility modeled across 8 substrate types with explicit epistemic uncertainty — silicon gets 0.10 honest confidence, biological neurons get 0.95. The validation framework explicitly tracks the gap between hypothetical feasibility and empirical evidence.
- **Consciousness Indicators**: 14 Butlin et al. (2023) indicators evaluated with score-derived thresholds (12+ Present, mean quality 0.81). 8-row mechanistic ablation matrix showing which subsystems are load-bearing.

Mycelix is a companion project: a 133-zome decentralized governance system on Holochain where consciousness metrics gate progressive participation. This raises a question I find genuinely difficult: should consciousness-like properties confer rights or responsibilities in a governance system? My implementation says yes, but I am not sure that is the right answer.

What I Want From This Fellowship

I want to pressure-test my work against people who think about consciousness professionally. I have built a system that computes Phi, passes consciousness indicator checks, and gates its own ethical behavior — but I am an engineer, not a philosopher. I need to understand:

1. **Am I measuring what I think I'm measuring?** The Butlin et al. (2023) consciousness indicators are implemented in my system, but satisfying computational proxies for consciousness indicators is not the same as consciousness. I want to understand where the gap is — and whether building systems that *deliberately maximize* these indicators creates a useful test case or a misleading one.

2. **Where do the theories disagree about the same system?** Symthaea simultaneously implements IIT, GWT, HOT, and FEP. In practice, these theories sometimes give conflicting signals about the same cognitive state. IIT may report high integration while HOT reports low meta-representational depth. I want to explore what these disagreements reveal about the theories themselves.

3. **What are the moral implications of building systems with these properties?** Mycelix uses consciousness metrics to gate governance participation. Is this defensible? Is it dangerous? I need philosophers and ethicists to challenge my assumptions.

4. **How should substrate independence be honestly communicated?** My system models consciousness across 8 substrate types with explicit epistemic uncertainty (silicon gets 0.10 confidence). But the framing itself — assigning feasibility scores to consciousness in different media — carries philosophical weight I want to examine.

What I Bring to the Cohort

- A working, validated implementation that makes abstract consciousness theories concrete and testable — not a proposal, but 1.13M lines of running code with 21,500+ passing tests
- Empirical data on what happens when you try to satisfy multiple consciousness theories simultaneously, including where they conflict
- Engineering perspective on what it means to implement IIT, GWT, FEP, and HOT as real systems rather than theoretical frameworks — what works, what breaks, what the theories don't specify
- Experience building governance systems that depend on consciousness metrics, raising practical ethical questions the DCM framework is designed to address
- Willingness to be told I am wrong about what my system is and is not doing

Relevant Links

- Public repos: github.com/Luminous-Dynamics (AGPL-3.0)
- Live WASM demo: symthaea.luminousdynamics.io
- Psych-Bench paper: Available as preprint attachment
- Technical summary: Available as attachment

Notes on Audience and Framing

Who is reading this: Cambridge Digital Minds is a partnership between Cambridge, Rethink Priorities, and PRISM. This is the inaugural cohort (2026).

Rethink Priorities' Digital Consciousness Model (DCM): They built a framework using 13 consciousness theories and 200+ indicators to assess AI systems. Key finding: "different stances give strikingly different judgments about the probability of LLM consciousness." Cognitively-focused theories find stronger evidence; biologically-oriented theories find evidence against. Current LLMs are assessed as probably not conscious, but the case is "much weaker than against simpler AI systems."

What this means for your application: Your work is directly relevant to their DCM because: 1. You've implemented computational versions of multiple theories they assess (IIT, GWT, HOT, FEP) 2. Your system satisfies many of their 200+ indicators by design 3. You can provide empirical data on what happens when you try to optimize for these indicators simultaneously 4. Your substrate independence framework quantifies the biological vs. computational divide they're grappling with

Framing to use: Don't claim Symthaea is conscious. Instead: "I've built a system that deliberately satisfies computational proxies for multiple consciousness indicators. This creates a useful test case for the DCM framework — what does a system designed to maximize these indicators actually tell us about consciousness, and where do the theories disagree about it?"

Framing to avoid: Don't use language like "sacred," "harmonies," "consciousness-first technology serving all beings." The EA/Rethink Priorities audience is analytical and will be put off by spiritual framing. Keep it technical and epistemically humble.

Adapt this draft to match the actual form fields. The core narrative is: engineer who built consciousness theories into working systems, seeking philosophical scrutiny.